

INDEPENDENT UNDERGRADUATE RESEARCH MANUSCRIPT

A Risk-Aware Predictive Maintenance Decision Framework

From Statistical Process Control and Nonlinear Failure Prediction to Constrained Operating Decisions

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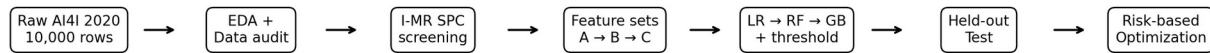
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Abstract

Predictive maintenance is often framed as a binary classification task, but an engineering decision system must do more than produce a high discrimination score. It must identify whether a process signal is operationally meaningful, preserve a defensible validation sequence, expose the trade-off between missed failures and false alarms, and convert a locked risk model into feasible actions. This independent undergraduate study develops an auditable workflow that links exploratory data analysis (EDA), individuals-moving range (I-MR) statistical process control (SPC), physically motivated feature engineering, imbalanced classification, threshold selection, failure-mode diagnosis, constrained operating-point search, and a small linear-programming extension. The analysis uses the 10,000-row AI4I 2020 synthetic predictive-maintenance dataset. Observation order is preserved in a contiguous 70/15/15 train-validation-test split. Mechanical SPC alarms were hypothesized to concentrate failure risk, while temperature variables were expected to require caution if their baseline drifted. Logistic regression was used as an interpretable linear baseline, Random Forest as a bagging-based nonlinear benchmark, and Gradient Boosting as a sequential error-correcting ensemble. The final Gradient Boosting model and a failure-sensitive threshold of 0.65 were locked on validation data before the test set was opened. On the held-out test block, precision was 0.5526, recall 0.7241, F1 0.6269, ROC-AUC 0.9615, and PR-AUC 0.7463, detecting 21 of 29 failures. Torque and mechanical-power SPC alarms were selective but had low coverage, whereas temperature-difference alarms were dominated by a level shift. Feature engineering materially improved failure ranking, while SPC alarm variables contributed little incremental importance once continuous variables were available. The final model's first weak learner recovered split points near documented synthetic failure thresholds, including 1386.5 rpm, 8.65 K, 3.50 kW, and 201.5 min. A constrained 60×60 RPM-torque search yielded a feasible low-score operating candidate at 1768.339 rpm and 41.446 Nm (7.675 kW; model score 1.3046%). These outputs are treated as a reproducible decision prototype, not as calibrated field risk or causal operating policy.

Keywords: predictive maintenance; statistical process control; imbalanced classification; random forest; gradient boosting; process optimization; AI4I 2020

Research Logic: Hypothesis → Validation → Decision



H1-H2: Can SPC separate selective mechanical risk from baseline drift features and nonlinear ensembles help core failure risk types be translated into feasible decisions?

Figure 1. Research logic used in the study: hypotheses are tested sequentially before model outputs are translated into decisions.

1. Introduction

Unexpected equipment failure simultaneously affects throughput, quality cost, delivery reliability, and maintenance capacity. A predictive-maintenance model is therefore useful only when its statistical output can be interpreted within a process-monitoring and decision context. SPC and machine learning address different parts of this problem. SPC distinguishes common-cause and special-cause variation with transparent limits, whereas supervised learning can combine several measurements and nonlinear interactions to rank individual observations. Optimization can then translate those scores into candidate operating or maintenance actions. The central idea of this study is not that one technique replaces the others, but that each method should be assigned a role that can be audited.

The research began from a practical question: if sensor data contain both ordinary process variation and failure-related patterns, can interpretable SPC signals first reveal abnormal operation, can those signals then improve a failure classifier, and can the resulting failure score be used to support a constrained decision? This sequence was deliberately retained even when some intermediate expectations were not confirmed. In particular, a negative result—SPC alarms adding little incremental predictive importance—was considered informative because it clarifies the boundary between process monitoring and failure classification.

1.1 Research questions

RQ1 (process monitoring): Which operating variables are associated with failures, and can I-MR control charts identify selective abnormal-process signals rather than merely broad distribution shifts?

RQ2 (failure prediction): Do physically engineered variables and SPC-derived indicators improve failure-sensitive classification beyond original process variables, and do nonlinear ensembles outperform a linear baseline?

RQ3 (decision support): Once a model and threshold are locked, how can the resulting score be translated into feasible operating-point and planning examples without interpreting the score as a causal probability?

1.2 Pre-analysis hypotheses and decision criteria

Hypothesis	Expectation	Evaluation rule
H1	Mechanical SPC alarms (torque, power, speed) will have failure risk above the base rate, although coverage may be low.	Operational selectivity, risk lift > 1
H2	Temperature-related SPC may be unreliable under a drifting baseline.	High alarm rate with weak risk lift or visible level shift
H3	Physical feature engineering will improve ranking over the base feature set.	Model B PR-AUC / recall > Model A
H4	SPC alarms may add incremental information beyond engineered continuous features.	Model C > Model B, especially recall
H5	Tree ensembles will outperform logistic regression if failure mechanisms are threshold- and interaction-driven.	RF/GB PR-AUC and error counts
H6	A locked risk model can generate feasible decision candidates, but not a causal optimum without real cost/intervention data.	Constrained search + explicit caveats

2. Data and Study Design

2.1 AI4I 2020 dataset and known synthetic mechanisms

The AI4I 2020 dataset contains 10,000 observations and six primary process features: product quality Type (L/M/H), air temperature, process temperature, rotational speed, torque, and tool wear [1,2]. It is synthetic but designed to resemble industrial predictive-maintenance data. The public documentation also describes five synthetic failure mechanisms. These documented rules were not used as predictive inputs in this study; they are used only for post-hoc interpretation of whether the learned model recovers plausible structure.

Mode	Documented generation logic	Main variables
TWF	Tool replacement/failure occurs in a high-wear band	Tool wear

Mode	Documented generation logic	Main variables
	around 200-240 min, with stochastic replacement/failure behavior.	
HDF	Process-air temperature difference below 8.6 K AND rotational speed below 1380 rpm.	ΔT , RPM
PWF	Mechanical power below 3.5 kW OR above 9 kW.	Torque $\times$ RPM / power
OSF	Tool wear $\times$ torque exceeds type-dependent threshold (L/M/H).	Wear, torque, Type
RNF	Random failure component independent of process parameters.	Not predictable from sensors by construction

The distributed CSV analyzed here contains 339 overall machine failures (3.39%), creating a severe class-imbalance problem. Simple accuracy is therefore insufficient: a classifier predicting nearly every observation as normal can still appear strong. Precision, recall, F1, ROC-AUC, PR-AUC, and the raw confusion-matrix counts were evaluated, with PR-AUC and failure recall given particular weight [8-10].

2.2 Data-quality audit

Audit item	Observed value
Rows / columns	10,000 / 14
Missing values	0
Fully duplicated rows	0
Machine failure = 1	339
Rows with ≥ 1 failure-mode flag	348
Mode positive but Machine failure = 0	18 (all associated with RNF)
Machine failure = 1 but no mode flag	9
Total label-consistency exceptions	27

The 27 inconsistencies were not deleted or manually relabeled. Machine failure was preserved as the supervised target, while TWF/HDF/PWF/OSF/RNF were excluded from predictive inputs to avoid target leakage. Failure-mode indicators were used only after final test evaluation for diagnostic interpretation. This decision preserves the distributed dataset and makes the sensitivity issue explicit rather than silently correcting labels.

2.3 Sequential partition and feature engineering

Block	Observation order	n	Failures	Failure rate
Training	1-7,000	7,000	278	3.971%
Validation	7,001-8,500	1,500	32	2.133%
Test	8,501-10,000	1,500	29	1.933%

No real timestamp or asset identifier is provided. Row order was therefore treated as pseudo-time, and a contiguous split was chosen instead of a random split. This avoids mixing nearby future observations into training, but it does not reproduce a true longitudinal asset history. The declining failure prevalence across blocks also means that a single split is an important uncertainty source.

$$\text{Temperature difference [K]} = \text{Process temperature [K]} - \text{Air temperature [K]}$$

$$\text{Mechanical power [kW]} = \text{Torque [Nm]} \times 2\pi \times \text{Rotational speed [rpm]} / (60 \times 1000)$$

Mechanical power was introduced as a domain-informed interaction feature because torque and speed jointly define shaft power. EDA showed that failure observations had higher mean torque (50.17 vs 39.63 Nm), higher tool wear (143.78 vs 106.69 min),

lower mean speed (1496.5 vs 1540.3 rpm), and higher mechanical power (7.283 vs 6.245 kW). Torque and power were also strongly related, so later coefficient and importance analyses were interpreted cautiously rather than causally.

3. Statistical Process Control: Why It Was Used and What It Actually Contributed

3.1 Rationale for I-MR charts

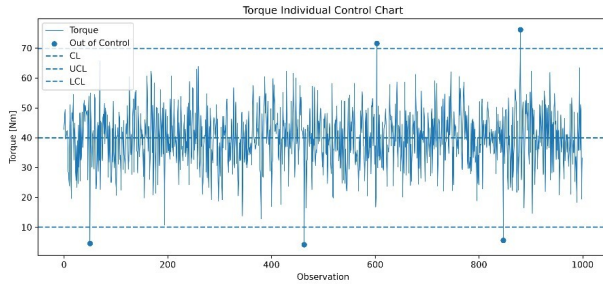
SPC was introduced before machine learning for two reasons. First, each row is an individual observation rather than a subgroup, making I-MR charts a natural univariate monitoring tool [3,4]. Second, the study deliberately tested whether interpretable special-cause flags could become useful model features. This was a hypothesis to be tested, not an assumption that SPC alarms must improve prediction.

$$MR_i = |x_i - x_{i-1}|; \quad \hat{\sigma} = MR / d_2, \quad d_2 = 1.128$$

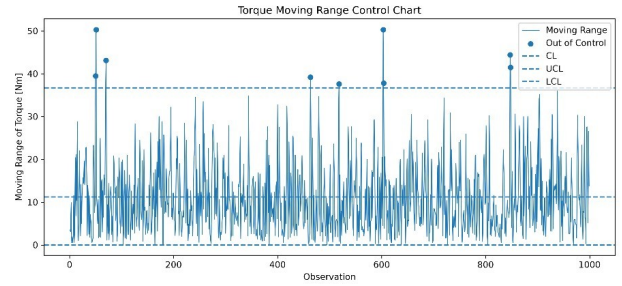
$$I\text{-chart: } CL = \bar{\bar{x}}, \quad UCL = \bar{\bar{x}} + 3\hat{\sigma}, \quad LCL = \bar{\bar{x}} - 3\hat{\sigma}$$

$$MR\text{-chart: } CL = MR\bar, \quad UCL = D_4 MR\bar, \quad LCL = D_3 MR\bar; \quad D_4 = 3.267, \quad D_3 = 0$$

All control limits were estimated from the 7,000-row training block and exported unchanged to later blocks. For torque, $CL=39.93$ Nm, $UCL=69.79$ Nm, and $LCL=10.06$ Nm. The first 1,000 observations were plotted for readability, but the limits themselves were estimated from all 7,000 training observations.



(a) Torque individual chart.



(b) Torque moving-range chart.

Figure 2. Torque I-MR monitoring. Out-of-control observations indicate special-cause behavior, not automatically a machine failure.

3.2 SPC screening results and the unexpected temperature result

Validation signal	Alarm %	Failure recall %	Failure rate in alarms %	False alarm %	Risk lift
Air temperature	77.07	65.62	1.82	77.32	0.85
Process temperature	91.13	96.88	2.27	91.01	1.06
Rotational speed	3.80	18.75	10.53	3.47	4.93
Torque	1.33	12.50	20.00	1.09	9.38
Mechanical power	1.47	12.50	18.18	1.23	8.52
Temperature difference	96.13	93.75	2.08	96.19	0.98

The expected mechanical pattern was partly confirmed. Torque and power alarms were rare but highly selective: their alarm groups had failure rates 20.0% and 18.18%, yielding risk lifts of 9.38 and 8.52. However, they covered only 12.5% of validation failures. Rotational speed offered broader but weaker coverage. Thus SPC produced escalation signals, not a comprehensive failure detector.

The temperature result was the most important negative finding. Temperature-difference alarms fired for 96.13% of validation observations, while risk lift was approximately 0.98. Instead of representing repeated isolated faults, the sequence plot showed a clear level shift: mean temperature difference increased from 9.610 K in training to 10.924 K in validation. A fixed-center chart therefore violated the practical baseline-stability assumption. Temperature-related SPC alarms were excluded from the final SPC feature set, while the continuous temperature variables remained available to machine learning.

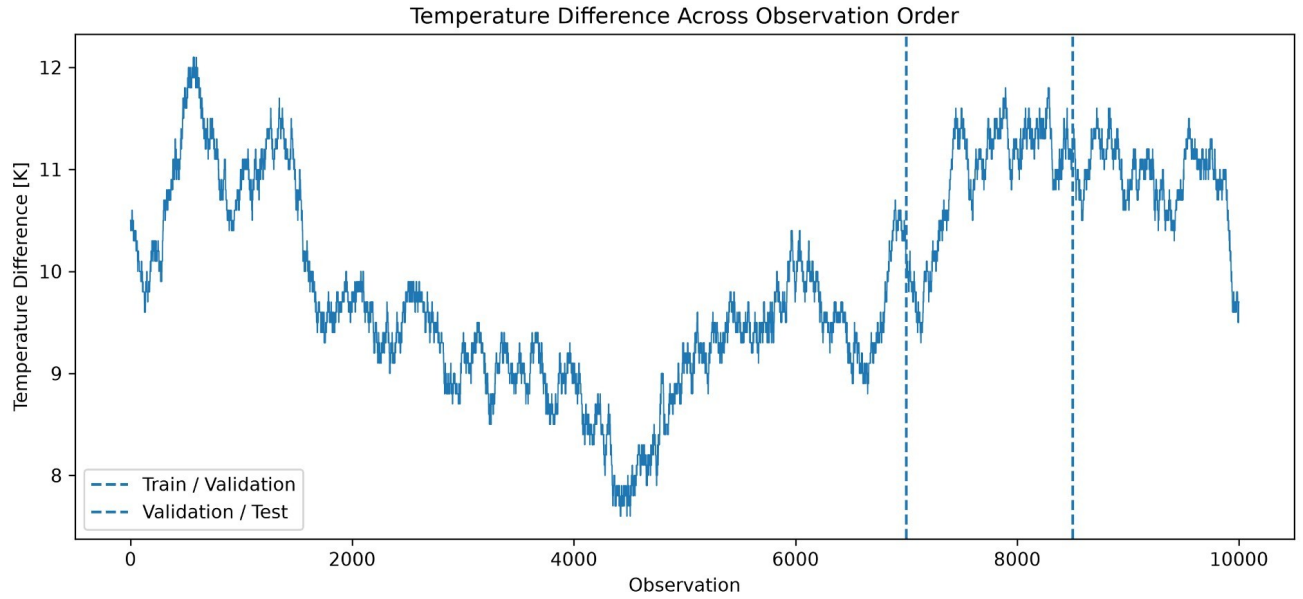


Figure 3. Temperature difference by observation order. The vertical lines mark the train/validation and validation/test boundaries. The level shift explains why a fixed SPC center generated near-continuous alarms.

3.3 Why SPC was still carried into Model C

Despite low coverage, the mechanical alarms were retained as candidate binary descriptors because they were operationally interpretable and had high conditional risk lift. The key question became whether the model could benefit from knowing both the continuous measurement and whether that measurement crossed an interpretable process-monitoring boundary. This decision created Model C and preserved a direct bridge from quality-control reasoning to supervised learning. The subsequent finding that SPC flags had near-zero permutation importance is therefore not a design failure; it is the empirical answer to H4.

4. Machine-Learning Methodology and Algorithmic Detail

4.1 Feature-set experiment: Models A, B, and C

Feature set	Inputs	Research purpose
Model A - Base	Type; air/process temperature; rotational speed; torque; tool wear	Can original process variables alone discriminate failure?
Model B - Engineered	Model A + temperature difference + mechanical power	Do physically motivated interactions improve ranking?
Model C - SPC Enhanced	Model B + torque, power, and RPM SPC alarms	Do interpretable special-cause indicators add incremental information?

The $A \rightarrow B \rightarrow C$ design is a nested feature experiment. Model C does not replace Models A and B; it contains them. Logistic regression was first held constant across A/B/C so that any performance change could be attributed primarily to feature representation rather than to a change in algorithm.

4.2 Logistic regression: interpretable linear baseline

$$\text{logit}(p) = \log[p/(1-p)] = \beta_0 + \sum_j \beta_j x_j; \quad p = 1/(1+e^{-z})$$

Logistic regression was selected as the first classifier because it provides a transparent linear probability model and a clear baseline for testing whether failure can be represented by an additive decision boundary. Numeric variables were standardized; Type was one-hot encoded. Because failures were rare, `class_weight='balanced'` was used so that positive observations contributed more strongly to the weighted log-loss. The implementation used `max_iter=2000` and `random_state=42`.

The expectation was not that logistic regression would necessarily win, but that it would reveal whether the engineered variables moved the classification problem in the expected direction. Model A achieved PR-AUC 0.3595 and recall 0.5625. Adding temperature difference and mechanical power in Model B increased PR-AUC to 0.4480 and recall to 0.6562. Model C increased

recall again to 0.6875 but PR-AUC slightly decreased to 0.4458. Thus H3 was supported, while H4 was only partially supported: SPC flags improved sensitivity at the default threshold but added little ranking information.

Logistic set	Precision	Recall	F1	ROC-AUC	PR-AUC	FP	FN	TP
A Base	0.2769	0.5625	0.3711	0.8855	0.3595	47	14	18
B Engineered	0.2386	0.6562	0.3500	0.9007	0.4480	67	11	21
C SPC Enhanced	0.2500	0.6875	0.3667	0.9023	0.4458	66	10	22

Model C coefficients were useful diagnostically but not causally. Standardized torque had a large positive coefficient (+7.454), speed was positive (+2.301), tool wear was positive (+0.894), and mechanical power was negative (-4.420). Because power is mathematically constructed from speed and torque, these coefficients are affected by strong dependence among predictors; the apparent signs cannot be read as physical causal effects. An ablation test was therefore more informative: removing power reduced PR-AUC from 0.4458 to 0.3079, while replacing torque and speed with power alone reduced PR-AUC further to 0.1785. Mechanical power complemented rather than replaced its source variables.

4.3 Random Forest: bagging, random feature selection, and nonlinear interactions

Random Forest was introduced after the logistic experiment to test a specific modeling hypothesis: if failure regions are governed by thresholds and interactions rather than a single linear boundary, an ensemble of decision trees should substantially improve discrimination [5]. Each tree is trained on a bootstrap sample; at each split, only a random subset of predictors is considered; terminal-node votes/probabilities are then aggregated across trees. This parallel bagging structure reduces the variance of a single deep tree while retaining the ability to model rules such as 'low speed AND low temperature difference' or 'high wear AND high torque'.

$$\hat{p}_{RF}(x) = (1/B) \sum_{b=1}^B \hat{p}_b(x), \quad B = 300 \text{ trees}$$

The Random Forest used 300 trees, class_weight='balanced', random_state=42, and all Model C features. Scaling was unnecessary because tree splits depend on order and thresholds rather than Euclidean magnitude. On validation data, precision rose to 0.7500, F1 to 0.7000, and PR-AUC to 0.7455, while false positives fell from 66 (Logistic C) to 7. Recall was 0.6562. This sharp gain strongly supported the nonlinearity/interaction hypothesis.

4.4 Gradient Boosting: sequential error correction

Gradient Boosting was selected as the third model because it attacks the problem differently from Random Forest. Rather than fitting many trees independently and averaging them, boosting adds shallow trees sequentially so that each new learner reduces the residual loss left by the current ensemble [6]. This was expected to be useful for rare failure patterns that a previous tree handled poorly. The final implementation used 200 estimators, learning_rate=0.05, max_depth=3, and random_state=42. Balanced sample weights were supplied during fitting to reduce majority-class dominance.

$$F_m(x) = F_{m-1}(x) + \eta h_m(x), \quad \eta = 0.05, \quad m = 1, \dots, 200$$

Why Three Models? Increasing Representational Complexity

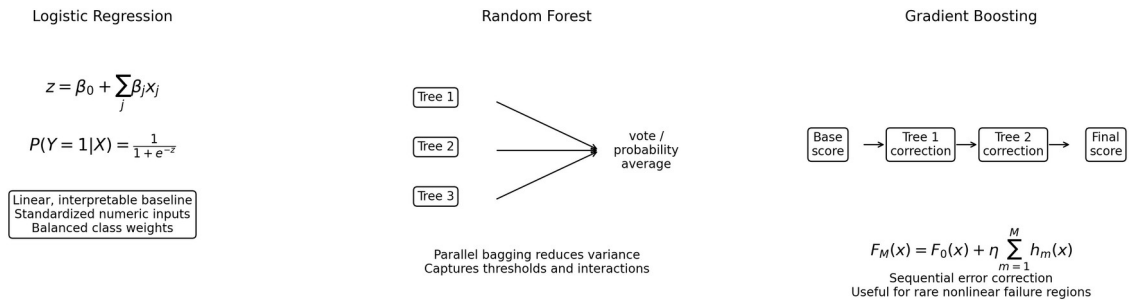


Figure 4. Modeling rationale. Logistic regression tests a linear additive boundary; Random Forest tests bagged nonlinear thresholds; Gradient Boosting sequentially corrects difficult residual patterns.

4.5 What a boosting tree actually learned

The serialized final model permits inspection of its first depth-3 weak learner. Importantly, the node values in Figure 5 are residual corrections, not calibrated failure probabilities. Nevertheless, the split thresholds are highly informative. The first tree split at 1386.5 rpm, then 8.65 K temperature difference; another branch split at 3.496 kW, and a later branch at 201.5 min tool wear. These

boundaries closely echo the public synthetic-generation rules for HDF, low-power PWF, and high-wear failure regions [1]. The ensemble subsequently adds 199 more trees, so the final classifier cannot be reduced to this single tree, but the first iteration demonstrates why a nonlinear tree model can fit this dataset efficiently.

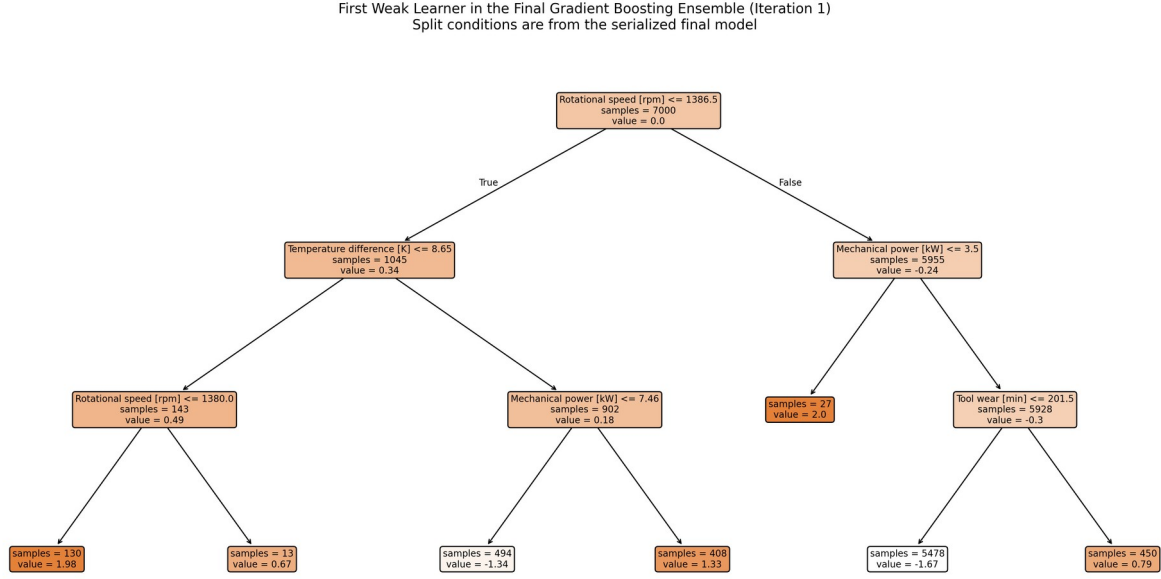


Figure 5. First weak learner extracted from the serialized final Gradient Boosting model. Node values are boosting residual corrections; split thresholds are exact values from the stored model.

Mechanism	Public synthetic rule	Observed first-tree split
HDF	$\Delta T < 8.6$ K and RPM < 1380	Tree 1: RPM ≤ 1386.5 ; $\Delta T \leq 8.65$; next split RPM ≤ 1380
PWF (low)	Power < 3.5 kW	Tree 1: power ≤ 3.496 kW
High-wear region	TWF documented around 200-240 min; OSF also depends on wear $\times$ torque	Tree 1: tool wear ≤ 201.5 min
High-load region	PWF upper limit > 9 kW; OSF wear $\times$ torque threshold	Tree 1 uses power 7.46 kW in the low-RPM branch; later ensemble trees refine the boundary

4.6 Evaluation metrics and threshold policy

$$\text{Precision} = TP/(TP+FP); \text{ Recall} = TP/(TP+FN); F1 = 2 \cdot \text{Precision} \cdot \text{Recall} / (\text{Precision} + \text{Recall})$$

Accuracy was reported but not used as the primary selection criterion because only 3.39% of observations are failures. ROC-AUC summarizes ranking across thresholds but can look strong under class imbalance; PR-AUC more directly reflects the positive-class precision-recall trade-off [8-10]. Raw FP, FN, TP, and TN counts were therefore reported alongside all summary metrics.

The probability threshold was treated as an operating choice rather than a fixed property of the classifier. Random Forest and Gradient Boosting thresholds from 0.10 to 0.90 were screened on validation data. The final policy favored recall because a missed failure was judged more consequential than an additional inspection. Gradient Boosting at threshold 0.65 achieved validation precision 0.5435, recall 0.7812, and F1 0.6410. This model and threshold were frozen before the test block was evaluated.

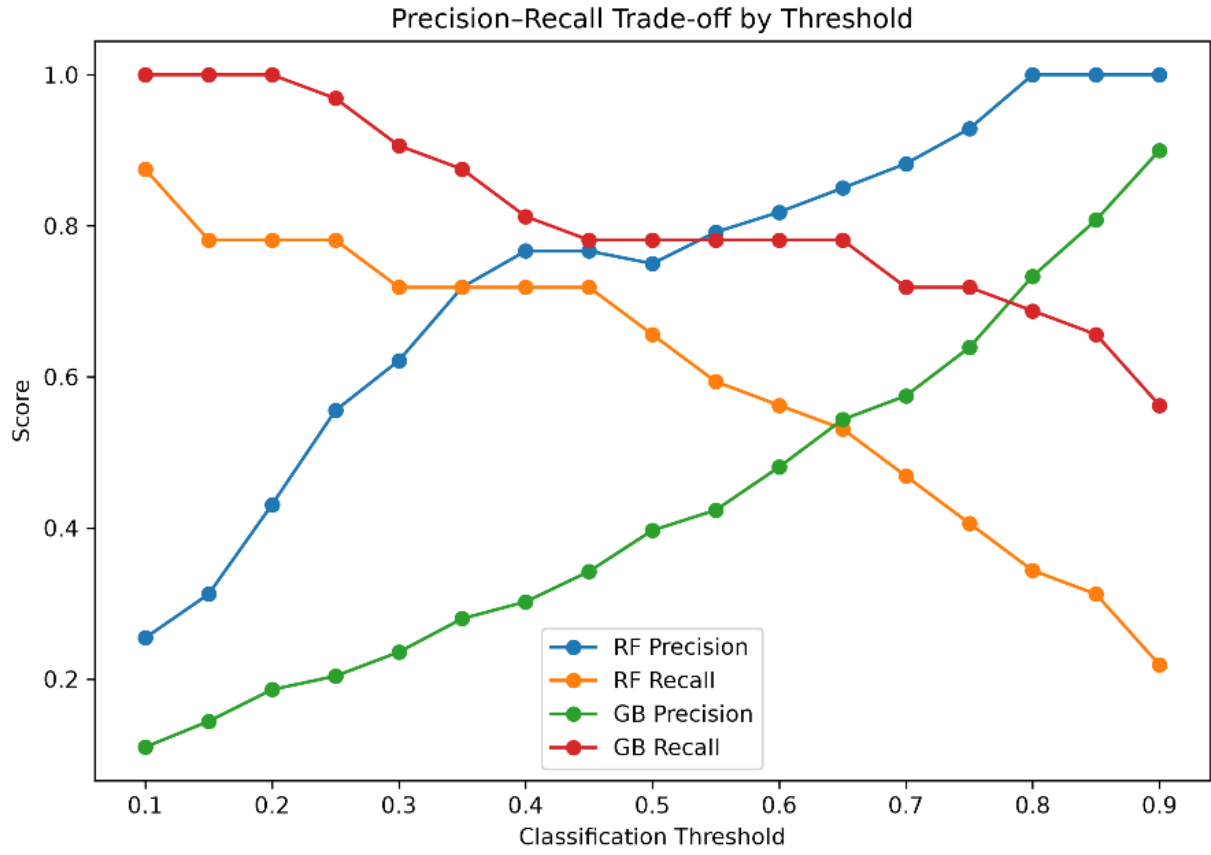


Figure 6. Precision-recall trade-off as the classification threshold changes. The selected Gradient Boosting threshold (0.65) was determined from validation data, not from the test block.

5. Results: Prediction, Interpretation, and Failure-Mode Diagnosis

5.1 Cross-algorithm validation

Model	Precision	Recall	F1	ROC-AUC	PR-AUC	FP	FN	TP
Logistic C	0.2500	0.6875	0.3667	0.9023	0.4458	66	10	22
Random Forest	0.7500	0.6562	0.7000	0.9736	0.7455	7	11	21
Gradient Boosting	0.3968	0.7812	0.5263	0.9857	0.7718	38	7	25

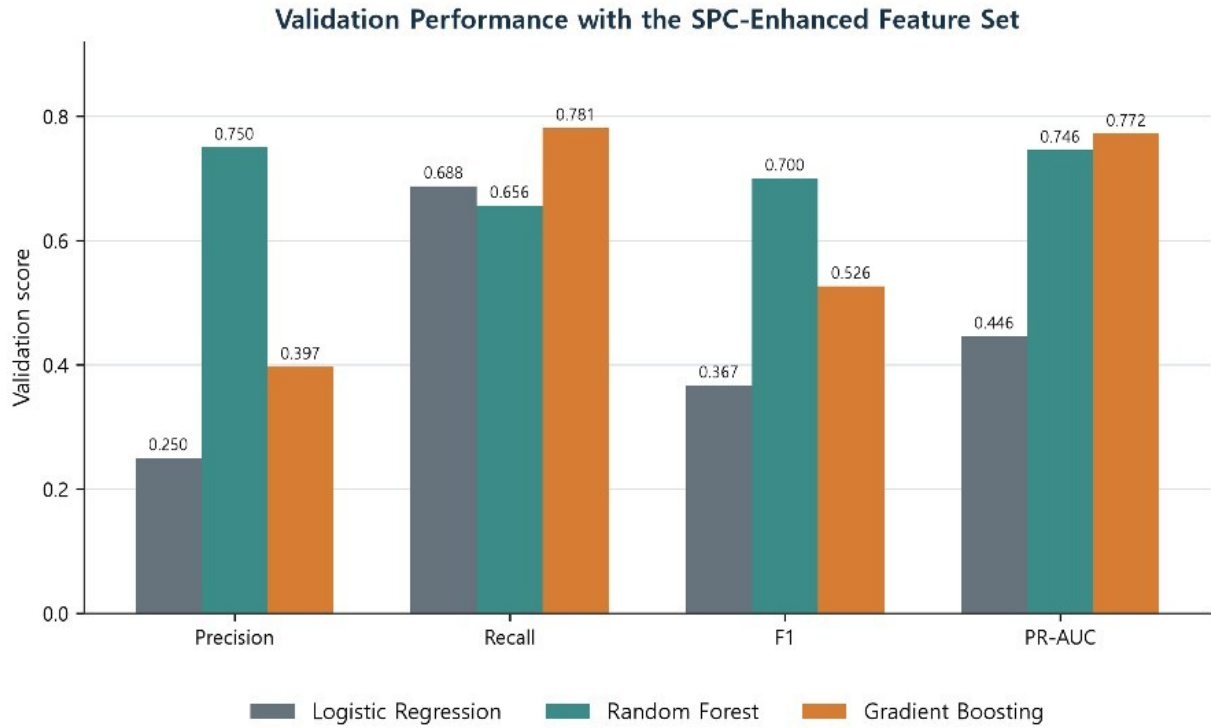
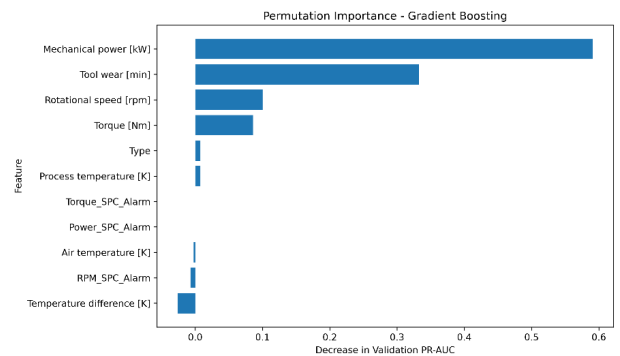
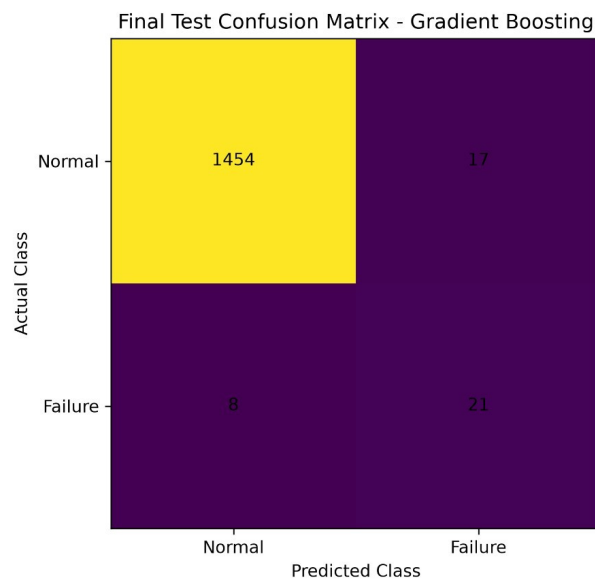


Figure 7. Validation comparison on the same SPC-enhanced feature set. Random Forest maximized precision/F1, whereas Gradient Boosting produced the highest recall and PR-AUC.

Random Forest would be the stronger choice if false alarms were the dominant operational cost. Gradient Boosting was selected because the project objective prioritized failure coverage: it detected 25 of 32 validation failures at the default threshold and retained the same 0.7812 recall at the selected 0.65 operating point while increasing precision. This is an explicit value judgment, not a universal claim that Gradient Boosting is always superior.

5.2 Locked test evaluation

Block	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC	FP	FN	TP	TN
Validation, threshold 0.65	0.9813	0.5435	0.7812	0.6410	0.9857	0.7718	21	7	25	1447
Held-out test	0.9833	0.5526	0.7241	0.6269	0.9615	0.7463	17	8	21	1454



(b) Validation PR-AUC permutation importance.

(a) Held-out test confusion matrix.

Figure 8. Final evaluation and interpretation. The test block contains 29 failures; 21 were detected and 8 were missed.

The modest drop from validation recall 0.7812 to test recall 0.7241 and PR-AUC 0.7718 to 0.7463 indicates that the selected operating point generalized reasonably to the later block. The test result is not used to re-tune the model. This separation is central to the credibility of the study.

5.3 Feature importance: an expected and an unexpected result

Built-in Gradient Boosting importance ranked rotational speed (0.3467), mechanical power (0.2697), tool wear (0.2046), temperature difference (0.0905), and torque (0.0758) as the dominant variables. Validation permutation importance told a related but sharper story: mechanical power caused the largest PR-AUC decrease when permuted (0.5909), followed by tool wear (0.3325), speed (0.1001), and torque (0.0859). By contrast, torque and power SPC alarms had essentially zero permutation contribution, and RPM SPC alarm was slightly negative. This was contrary to the initial expectation that SPC flags might materially strengthen Model C.

The interpretation is not that SPC was useless. SPC and machine learning were answering different questions. A torque alarm answered 'is this observation statistically unusual relative to the training baseline?' and had a risk lift of 9.38. Gradient Boosting answered 'which combination of continuous values best ranks failure?' Once the raw continuous measurements were available, a coarse binary alarm added little new information. The negative incremental result therefore defines a useful division of labor: SPC for interpretable monitoring and escalation, ML for multivariate failure ranking.

5.4 Failure-mode diagnosis after model selection

Mode	Actual	Detected	Missed	Detection	Mean score	Power kW	Wear min	RPM	Torque
TWF	7	1	6	14.3%	0.325	5.801	210.4	1686	35.4
PWF	7	7	0	100.0%	0.978	9.548	115.1	1365	66.9
OSF	17	16	1	94.1%	0.905	8.236	207.6	1362	57.8

The final model was especially strong on PWF and OSF but weak on TWF. This pattern is consistent with the variables that dominate the ensemble. PWF is directly power-defined in the synthetic generator, and OSF depends on the product of wear and torque; both mechanisms therefore align strongly with the continuous mechanical signals used by the model. TWF, in contrast, includes stochastic replacement/failure behavior within a high-wear range, making tool wear alone insufficient. HDF and RNF had no machine-failure-positive examples in the test block and were not assigned a test detection rate.

6. From Prediction to Decision: Optimization

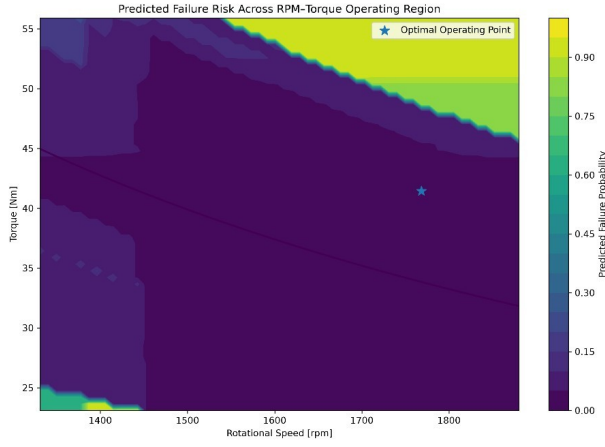
6.1 Why the operating-point problem is not a simplex problem

The final Gradient Boosting score is nonlinear and piecewise constant, and mechanical power is proportional to $\text{RPM} \times \text{torque}$. Therefore, the operating-point problem is not a linear program and should not be forced into simplex form. The decision variables were rotational speed and torque; surrounding temperature, product type, and tool wear were fixed at a representative scenario; candidate points were restricted to the 5th-95th percentile training range. A 60×60 grid provided 3,600 candidate operating points.

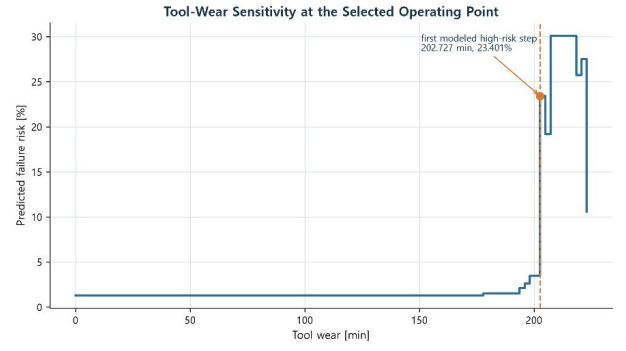
$$\text{minimize } \hat{p}_{\text{failure}}(\text{RPM}, \text{Torque} \mid \text{fixed scenario})$$

$$\text{subject to: } \text{Mechanical power} \geq \text{median training power}; \quad \text{RPM}_{0.05} \leq \text{RPM} \leq \text{RPM}_{0.95}; \quad \text{Torque}_{0.05} \leq \text{Torque} \leq \text{Torque}_{0.95}$$

Of the 3,600 candidates, 1,999 satisfied the power/SPC feasibility conditions. Several candidates shared the minimum model score. A representative low-score point was 1768.339 rpm and 41.446 Nm, corresponding to 7.675 kW and an uncalibrated model score of 1.3046%. This is a feasible candidate within the sampled grid, not a unique continuous optimum or experimentally verified safe setting.



(a) Conditional RPM-torque risk surface and selected point.



(b) Tool-wear sensitivity at the selected operating point.

Figure 9. Decision-layer diagnostics. Tree-based risk is highly nonlinear and shows abrupt transitions; the plotted percentages are model scores, not calibrated field failure probabilities.

6.2 Tool-wear sensitivity and maintenance interpretation

At the selected operating point, model score remained near 1.3% through low and moderate tool-wear values and then increased sharply in the high-wear region. The first of 100 sampled wear values crossing 5%, 10%, and 20% was approximately 202.727 min, where the score jumped to 23.401%. Because Gradient Boosting is tree-based, the response is stepwise and later high-wear values were not monotonic. The approximately 203-min transition should therefore be described as a model-based warning region, not a causal replacement threshold or safe-life estimate.

6.3 Linear-programming / simplex extension

A separate two-mode planning example was created to show how a locked model score could enter an operations-research decision. Mode A used the representative low-score operating point. Mode B used a high-output, high-score point (1880 rpm, 55.9 Nm, approximately 11.005 kW, score 92.635%). Let x_A and x_B be operating hours during an eight-hour shift. The illustrative objective maximized output subject to shift length and a scenario-defined risk-exposure budget.

$$\text{maximize } Z = P_A x_A + P_B x_B$$

$$\text{subject to } x_A + x_B \leq 8; \quad r_A x_A + r_B x_B \leq R_{\max}; \quad x_A, x_B \geq 0$$

The resulting LP returned 4.8 h in Mode A and 3.2 h in Mode B with an illustrative production score of 72.056. This solution is intentionally treated as a transfer demonstration rather than an empirical policy: the risk budget itself was constructed from the same 60/40 mixture, so the LP largely reproduces the scenario assumption. A real application would estimate maintenance capacity, downtime cost, production value, and expected failure loss from plant records before simplex or mixed-integer optimization is justified.

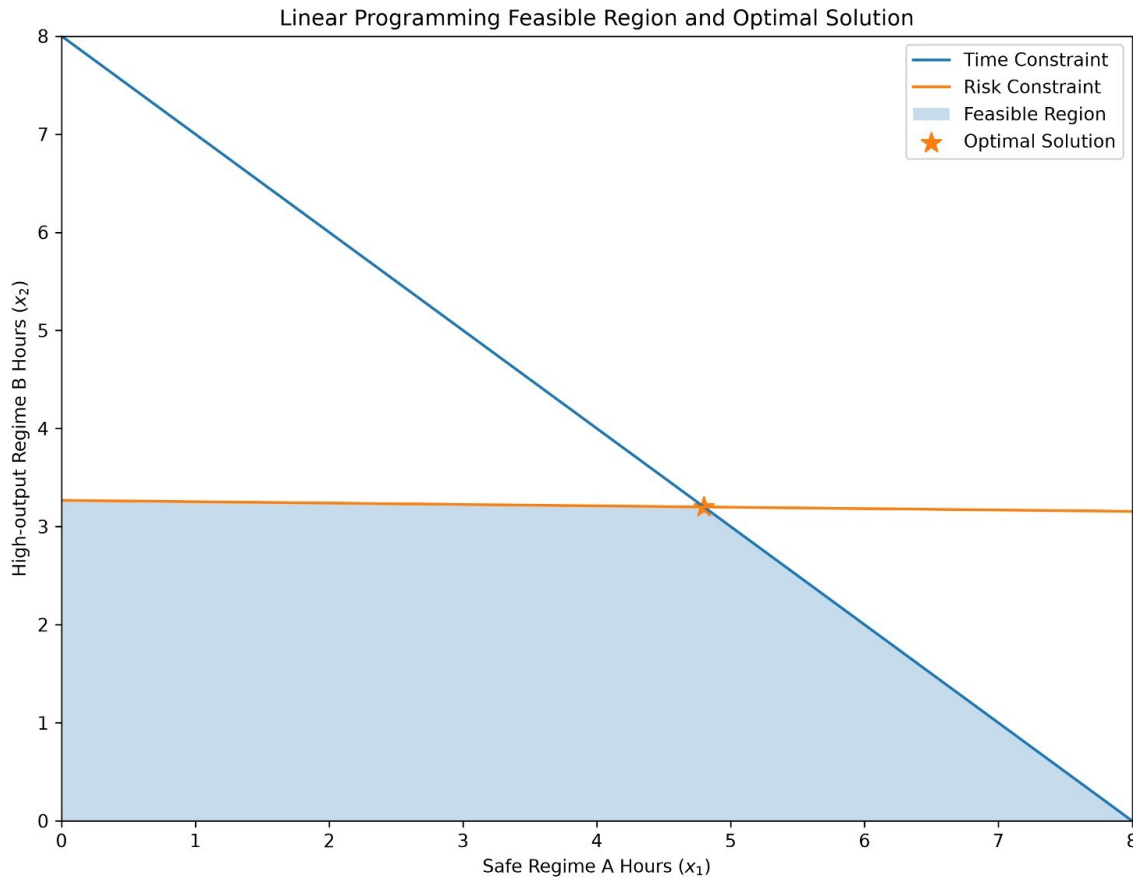


Figure 10. Feasible region and vertex solution for the illustrative two-mode linear program. The simplex principle is appropriate here because the planning objective and constraints are linear.

7. Discussion: What Was Learned from the Research Process

7.1 Hypothesis review

Hypothesis	Assessment	Evidence / interpretation
H1	Supported with qualification	Torque/power/speed alarms concentrated risk, but failure coverage was limited.
H2	Supported	Temperature SPC mainly captured a baseline level shift rather than selective failure events.
H3	Supported	Model B improved recall (0.5625 → 0.6562) and PR-AUC (0.3595 → 0.4480).
H4	Partially supported / mostly negative	Model C recall increased to 0.6875, but PR-AUC slightly decreased and SPC permutation importance was near zero.
H5	Supported	RF/GB greatly outperformed Logistic C in PR-AUC; GB achieved the highest validation recall and PR-AUC.
H6	Supported only as a prototype	Feasible candidate decisions were generated, but uncalibrated scores and assumed budgets prevent causal or economic policy claims.

7.2 Why the imperfect path is scientifically useful

The strongest contribution of this undergraduate study is not a new algorithm. It is the traceable decision process. SPC was used because it offered an interpretable way to distinguish ordinary from unusual variation and because the research hypothesis was that

those flags might enrich a classifier. The temperature experiment rejected a naive use of fixed control limits under drift. The Model C experiment then showed that even high-risk-lift alarms can provide little incremental ranking information when the underlying continuous variables are already available. This negative result prevented the final narrative from becoming 'SPC automatically improves ML' and led instead to a more defensible interpretation: SPC is a sparse monitoring layer, while nonlinear ML provides broader multivariate coverage.

Likewise, model choice was not a leaderboard exercise. Logistic regression established how far a transparent additive model could go and demonstrated the value of engineering mechanical power. Random Forest tested whether threshold interactions mattered and sharply reduced false alarms. Gradient Boosting then traded some precision for additional failure coverage and achieved the highest PR-AUC, fitting the failure-sensitive objective. The exact first-tree split values showed that the ensemble was not exploiting arbitrary noise: it immediately recovered thresholds close to the synthetic generator. At the same time, this alignment is also a warning that AI4I is easier and cleaner than real plant data.

7.3 Reliability of the workflow

Several design choices improve internal credibility: all transformations and SPC limits were based on the training block; the validation block was used for model and threshold choice; the threshold was locked before the test block was opened; and the final test result was not used to revise the selected model. Numerical findings are reproducible from the stored notebooks and model artifacts. However, reliability should not be confused with external validity. The single pseudo-sequential split, synthetic generator, and absence of asset-level repeated measurements limit generalization.

8. Limitations and Future Research

- Synthetic data: AI4I intentionally contains simplified, partly rule-driven failures and omits many characteristics of real maintenance data, including sensor aging, missingness, repair interventions, operating-context changes, censored lifetimes, and machine-to-machine heterogeneity.
- Pseudo-time rather than real time: UDI order was used as a sequence, but elapsed time, shifts, shutdowns, tool replacements, and asset identity are unavailable. The apparent temperature drift cannot be assigned to a physical day/night or heat-up mechanism without timestamps.
- Single validation path: one contiguous 70/15/15 split was used. Rolling-origin evaluation, repeated blocked validation, or cross-asset validation should replace a single split in future work.
- Uncalibrated probabilities: the Gradient Boosting outputs discriminate well but were not probability-calibrated. Optimization therefore minimizes a model score, not a validated absolute failure probability.
- Limited algorithm search: Logistic Regression, Random Forest, and Gradient Boosting were selected to test progressively richer assumptions, not to exhaustively benchmark all learners. Future work could include XGBoost/LightGBM, calibrated ensembles, cost-sensitive learning, survival analysis, and sequence models when true histories exist.
- SPC assumptions: I-MR charts were used as exploratory monitoring devices. Autocorrelation, nonstationarity, and distributional assumptions were not fully established. EWMA/CUSUM, adaptive limits, change-point detection, and multivariate T^2 /MEWMA methods are natural extensions.
- Causality: feature importance, tree splits, and risk surfaces identify predictive structure but do not prove that changing a variable will reduce failures. Prospective intervention or quasi-experimental data are needed before operating recommendations can be causal.
- Optimization inputs: production value, inspection capacity, maintenance duration, spare-part cost, and downtime loss were unavailable. The LP risk budget is scenario-based and cannot be interpreted as an economically optimal plan.

8.1 Proposed next-stage research program

Stage	Recommended extension
1. Real longitudinal data	Collect timestamp, asset ID, operating regime, failures, maintenance actions, component replacements, and downtime/cost.
2. Monitoring redesign	Estimate stable baselines by asset/regime; add EWMA/CUSUM/change-point detection; evaluate SPC alarm precision and delay.
3. Predictive validation	Rolling-origin and cross-asset validation; probability calibration; cost curves; subgroup robustness; missing-data stress tests.
4. Causal / intervention layer	Evaluate whether changing RPM/torque or replacing a worn tool actually

Stage	Recommended extension
	changes failure incidence.
5. Decision optimization	Use calibrated failure risk with real production and maintenance costs in stochastic, robust, or mixed-integer optimization.
6. Deployment study	Prospective shadow-mode monitoring, alarm review by engineers, then controlled decision-support deployment.

9. Conclusion

This study followed an intentionally transparent path from process monitoring to prediction and then to decision support. The path was not uniformly successful, which is precisely what made the final interpretation stronger. Mechanical SPC alarms were operationally meaningful but sparse; temperature SPC exposed a baseline shift; engineered physical features materially improved prediction; SPC flags provided little incremental importance once continuous measurements were available; and nonlinear tree ensembles captured failure structure far better than a linear baseline. A failure-sensitive Gradient Boosting model, fixed before test evaluation, detected 21 of 29 held-out failures with PR-AUC 0.7463. Inspection of the first boosting tree recovered thresholds close to documented synthetic failure rules, while post-hoc failure-mode results showed strong PWF/OSF coverage and weak TWF coverage. Finally, constrained search and a small LP demonstrated how a locked score can be moved toward engineering action without pretending that model scores are causal probabilities. The main lesson is therefore methodological: credible predictive maintenance requires a chain of auditable choices—baseline, hypothesis, monitoring signal, predictive model, threshold, error analysis, and feasible decision—not simply a high accuracy number.

Statements and Declarations

Research status. This manuscript reports independent undergraduate research by Doyun Shin, Department of Industrial Engineering, Gachon University. It is not presented as accepted, published, or peer-reviewed by an SCI/KCI journal.

Author contribution. The author formulated the research questions and hypotheses, selected the analytical sequence, executed and reviewed the analyses, interpreted the results, and assumes responsibility for the final manuscript and repository.

Generative AI assistance. Generative AI was used as an auxiliary tool for selected Python code drafting, debugging, explanation, language editing, and document-layout support. Numerical outputs, tables, model artifacts, and empirical figures were checked against the stored dataset and notebooks. AI assistance was not treated as an empirical source, and final analytical decisions and interpretations remain the author's responsibility.

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Data and code availability. The source dataset is the AI4I 2020 dataset from the UCI Machine Learning Repository (CC BY 4.0). The project repository contains the EDA, SPC, machine-learning, and optimization notebooks together with model and figure artifacts.

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Appendix A. Reproducibility Record

Item	Locked value / rule
Data	AI4I 2020 CSV, 10,000 rows
Split	Contiguous 70/15/15 by observation order
Target	Machine failure
Final features	Type; six original process variables; temperature difference; mechanical power; torque/power/RPM SPC alarms
Logistic	Balanced class weight; max_iter=2000
Random Forest	300 trees; balanced class weight; random_state=42
Gradient Boosting	200 estimators; learning_rate=0.05; max_depth=3; balanced sample weights
Final threshold	0.65 selected on validation before test
Test confusion matrix	TN=1454, FP=17, FN=8, TP=21
Grid search	RPM 5th-95th percentile × Torque 5th-95th percentile; 60×60
Representative candidate	1768.339 rpm; 41.446 Nm; 7.675 kW; score 1.3046%
Software	Python; pandas; NumPy; scikit-learn; SciPy; Matplotlib; joblib